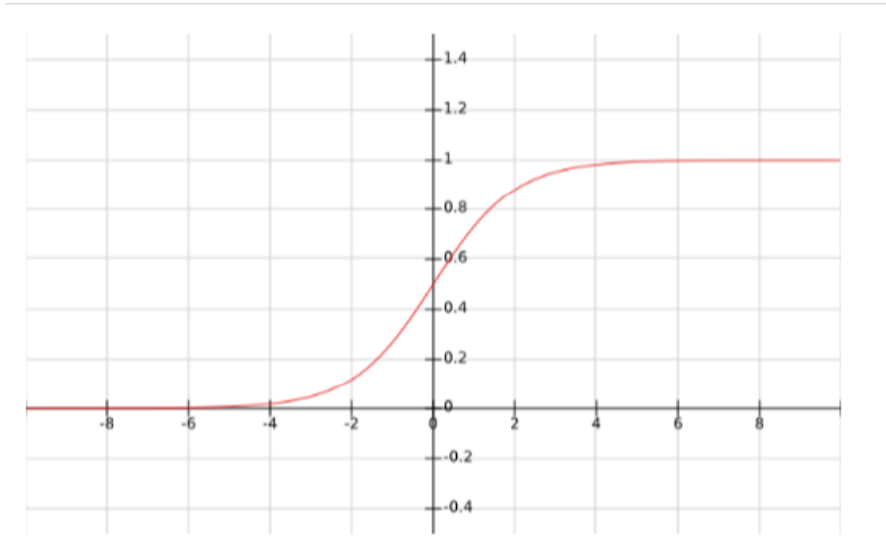


Sample Questions

Question 1. Let $\{1, 2, 3, 4\}$ represent the outcomes of a random experiment, and $P(1) = P(2) = P(3) = P(4) = 0.25$. Consider the following events: $X = \{1, 2\}$, $Y = \{2, 3\}$, $Z = \{3, 4\}$, $S = \{1, 2, 3\}$. Then, which of the following statements is/are true?

- (a) Z and S are independent
- (b) Y and S are not independent
- (c) X and Y are independent
- (d) X and Y are not independent

Question 2. What will be the value of the sigmoid function when $x = -\infty$ and $x = \infty$?



- (a) 1,0
- (b) $-\infty, \infty$
- (c) 0,1
- (d) $\infty, -\infty$

Question 3. A class contains 60% students who are incapable of changing their opinions about anything, and 40% of students are changing their minds at random, with probability 0.3 , between subsequent votes on the same issue. Then, the probability of a student randomly chosen voted twice in the same way is _____.

Question 4. Check whether the vectors $v_1 = \begin{bmatrix} -3 \\ 2 \\ -7 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 5 \\ -10 \\ -5 \end{bmatrix}$ are orthogonal. If orthogonal, find their corresponding orthonormal vectors.

- (a) Vectors v_1 and v_2 are orthogonal. The corresponding orthonormal vectors are $\begin{bmatrix} -0.25 \\ 0.17 \\ -0.58 \end{bmatrix}$ and $\begin{bmatrix} 0.25 \\ -0.5 \\ -0.25 \end{bmatrix}$.

(b) Vectors v_1 and v_2 are not orthogonal.

(c) Vectors v_1 and v_2 are collinear.

(d) Vectors v_1 and v_2 are orthogonal. The corresponding orthonormal vectors are $\begin{bmatrix} -0.38 \\ 0.25 \\ -0.89 \end{bmatrix}$ and

$$\begin{bmatrix} 0.41 \\ -0.82 \\ -0.41 \end{bmatrix}.$$

Question 5. Two machine learning models A and B are used separately for a problem to find whether a person has cancer or not. Based on the confusion matrix for Model A and confusion matrix for Model B,

Model A		Predicted		Model B		Predicted	
		Healthy	Cancerous			Healthy	Cancerous
Actual	Healthy	18	2	Actual	Healthy	7	3
	Cancerous	26	54		Cancerous	30	60

find which model is better in predicting the presence of cancer in a person.

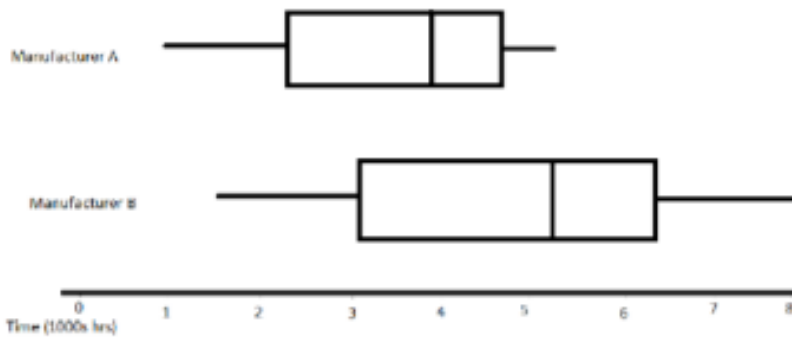
(a) The accuracy of both the models are close, hence any one of them can be chosen

(b) Model A

(c) Neither of the models can be chosen

(d) Model B

Question 6. The two box plots given below show the data collected by two light bulb manufacturers, A (top box plot) and B (bottom box plot) on the life span of light bulbs. From these box plots, find which of the following statement(s) is/are true.



I Manufacturer B produces light bulbs with better life span as the median is higher than that of Manufacturer A

II Manufacturer A produces light bulbs with better life span as the upper limit in Manufacturer A is equal to the median in Manufacturer B

Select the correct option.

- (a) I is False and II is True
- (b) Both I and II are True
- (c) I is True and II is False
- (d) Both I and II are False

Question 7. Two eigenvalues of 3×3 matrix X are $(1 + i)$ and 2. The determinant of the matrix X is

Question 8. Given a matrix $A_{3 \times 3} = \begin{bmatrix} 3a+4 & 6 & 2 \\ 1 & -3 & 5 \\ b & 8 & 4 \end{bmatrix}$ such that there exists at least one non-trivial vector in the null space of A. The value of $-156a + 36b$ is

- (a) 216
- (b) 432
- (c) 221
- (d) 108

Question 9. In logistic regression, if the posterior probability $Pr(Class = k|X = x)$ is linear in x then

- (a) decision boundary is linear
- (b) x is not related to Y
- (c) there is no decision boundary
- (d) decision boundary is non-linear

Question 10. Which among the following options best describes the behaviour of y for an increase in x for the given data?

x	y
6.37	10.47
-5.26	-3.48
1.48	4.61
-1.74	0.74
-4.35	-2.39
-1.44	1.1
-0.72	1.97

- (a) Non-monotonic
- (b) Monotonic increase
- (c) Monotonic decrease
- (d) None of the above

Question 11. What is $(I - A)^{-1}$ for a given matrix A , if $A^7 = 0$?

- (a) $I + A + A^2 + \dots + A^6$
- (b) Inverse is not guaranteed to exist

- (c) 0
- (d) A

Question 12. Let X be a 3×3 real matrix. The rank of X is two and the trace of the matrix X is 4. Then, the eigenvalues of X are:

- (a) -1,1,4
- (b) 2,2,0
- (c) 3,2,-1
- (d) 3,2,1

Question 13. The nullity of the matrix $M = \begin{bmatrix} 2 & 6 & -2 & -6 \\ 6 & 18 & -6 & -18 \\ 2 & 6 & -2 & -6 \\ 4 & 12 & -4 & -12 \end{bmatrix}$ is:

- (a) 2
- (b) 3
- (c) 0
- (d) 1

Question 14. Let $A = \begin{bmatrix} 1.0 & 1.0 \\ 1.32 & 1.0 \end{bmatrix}$ be a 2×2 matrix. Then the sum of the squares of the eigenvalues of the matrix A^6 is:

- (a) -1.0
- (b) 1486.55
- (c) 1013.86
- (d) 311.86
- (e) 1196.99

Question 15. X is an uniform random variable with support in $[-2, 2] \cup [99.5, 100.5]$. The mean of X is _____.

Question 16. Consider the following joint distribution of random variables X and Y :

$$f(x, y) = x(1 + 3y^2)/4, 0 < x < 2, 0 < y < 1$$

$$f(x, y) = 0, \text{ otherwise.}$$

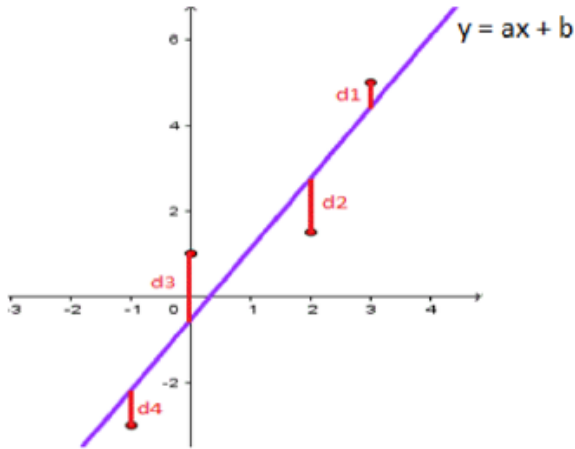
The marginal distribution of X is:

- (a) $x/4$
- (b) $y/4$
- (c) $x/2$
- (d) None of the above

Question 17. Consider a 4×4 matrix with every element being equal to 1. Its only non-zero eigenvalue is:

- (a) 4
- (b) 3
- (c) 1
- (d) 2

Question 18. In the given picture below, the line $y = ax + b$ is the linear regression line between x and y . d_1, d_2, d_3 and d_4 are the vertical deviations from each data point to the line. Find which of the following options is correct.



- (a) $d_1^2 + d_2^2 + d_3^2 + d_4^2$ should be maximum
- (b) $(d_1^2 + d_3^2)$ should be equal to $(d_2^2 + d_4^2)$
- (c) $d_1^2 + d_2^2 + d_3^2 + d_4^2$ should be equal to zero
- (d) $d_1^2 + d_2^2 + d_3^2 + d_4^2$ should be minimum

Question 19. Assume that the median for a right-skewed normal distribution is 1. Then the value of mean is:

- (a) Equal to one
- (b) Less than one
- (c) Equal to zero
- (d) Greater than one

Question 20. A controller is set up to control the outlet flow rate of a stirred-tank reactor. The desired flow rate is $14 \text{ m}^3/\text{s}$. The actual flow rate was then measured multiple times using a venturimeter to cross-check the performance of the controller. The following measurements were obtained from various trials:

$$x \text{ (m}^3/\text{s)} =$$

13.5	14.86	14.44	14.99	15.47	14.78
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With the help of hypothesis testing, determine whether the controller performance is good or bad. Assume that the true variance, $\sigma^2 = 0.97$ and significance, $\alpha = 0.01$

- (a) Accept null hypothesis, controller performance is good
- (b) Reject null hypothesis, controller performance is good
- (c) Reject null hypothesis, controller performance is bad
- (d) Accept null hypothesis, controller performance is bad

Question 21. Solve the given system of equations if a solution exists.

$$-2x - 2y - 2z = 6$$

$$x + y - 3z = 1$$

$$3x + y + 2z = -10$$

- (a) $x = -3, y = 1, z = -1$
- (b) No solution
- (c) $x = -5, y = 3, z = 1$
- (d) $x = -2, y = 0, z = -3$

Question 22. Find the mean and variance of the following normal distribution $f(x) = \sqrt{\frac{m}{2\pi KT}} e^{-\frac{mx^2}{KT}}$.

- (a) Mean = $\frac{1}{m}$, Variance = $\frac{kT}{m}$
- (b) Mean = x , Variance = kT
- (c) Mean = 0, Variance = $\frac{kT}{m}$
- (d) Mean = 0, Variance = kT

Question 23. A fair coin is tossed twice. What is the probability of obtaining at least one head?

- (a) 0.25
- (b) 0.50
- (c) 0.75
- (d) 1

Question 24. If $E[X] = 1$ and $Var(X) = 5$, $E[(2 + X)^2] = \underline{\hspace{2cm}}$.

Question 25. A binary classification model is fitted to determine the type of tumors from images of the tumors for 100 subjects with 10 negatives. The tumors are labeled either benign (class 1) or cancerous. The computed recall and precision for the fitted model are 8/9 and 8/9, respectively. Then, the false negative rate is $\underline{\hspace{2cm}}$.

Question 26. In the receiver operating curve (ROC), which of the following statements is TRUE for the point (1,1)?

- (a) The model classifies only the samples of the negative class correctly.
- (b) The model classifies only the samples of the positive class correctly.
- (c) The model does not classify samples of both negative and positive classes.
- (d) The model classifies samples of both negative and positive classes correctly.

Question 27. For the Leave-one-out-cross validation approach, which of the following statement(s) is/are TRUE?

- I. Leave-one-out-cross-validation approach leads to less bias in the mean-square-error estimation than the validation set approach
 - II. Leave-one-out-cross-validation approach leads to more bias in the mean-square-error estimation than the validation set approach
 - III. Leave-one-out-cross-validation approach will select a parsimonious model
 - IV. Leave-one-out-cross-validation approach may select a model with more parameters than the optimal one.
- Select the correct option.

- (a) I and II only
- (b) II and III only
- (c) II and IV only
- (d) I and IV

Question 28. If $p = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $q = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, then the linear combination $2p + 3q$ is -----.

- (a) $\begin{bmatrix} 11 \\ 16 \end{bmatrix}$
- (b) $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$
- (c) $\begin{bmatrix} 9 \\ 12 \end{bmatrix}$
- (d) $\begin{bmatrix} 16 \\ 11 \end{bmatrix}$

Question 29. Given that the two vectors $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ x \end{bmatrix}$ are known to be perpendicular, what is x ?

- (a) -1
- (b) 1
- (c) 2
- (d) -2

Question 30. What is the objective of the K-means clustering algorithm?

- (a) Maximizing within-cluster variance
- (b) Minimizing within-cluster variance
- (c) Maximizing between-cluster variance
- (d) Minimizing between-cluster variance

Question 31. The sum of eigenvalues of the matrix $A = \begin{bmatrix} -2 & -1 & 1 & 4 \\ -5 & 2 & 1 & 4 \\ -3 & -1 & 2 & 4 \\ 0 & -1 & 1 & 2 \end{bmatrix}$ is:

- (a) 5
- (b) 0
- (c) 4
- (d) 11

Question 32. Given a continuous and twice differentiable function of one variable, which of the following is a necessary condition for a minimum?

- (a) First derivative is positive
- (b) First derivative is negative
- (c) First derivative is zero
- (d) First derivative can take any real value

Question 33. Consider the following 3×3 matrix $A : \begin{bmatrix} x & y & 3 \\ 4 & 4 & 6 \\ 1 & 2 & 3 \end{bmatrix}$. The ordered pair(s) (x, y) for which $\det(A) = 0$ is/are:

- (a) (2,2)
- (b) (1,1)
- (c) (1,2)
- (d) (3,1)

Question 34. A short circuit tester is being used in a battery assembly line to test the batteries for short circuit. The battery will be rejected if the test is positive and the battery will be accepted if the test is negative. 93.69% of the non-defective batteries were accepted and 4.27% of the defective batteries were also accepted. In addition, 95.1% of the batteries are non-defective and 4.9% of the batteries are defective. If a battery is rejected, then the probability of the battery being non-defective is _____.

Question 35. The steps to implement the kNN algorithm are given below in random order. Choose the correct order required to implement the kNN algorithm

- i Order the labeled data points in the increasing order of the distance metric
 - ii Find the majority class of these k labeled data points and assign it to the test data point
 - iii Select the top k labeled data points and observe their class labels
 - iv Compute the distance metric between the test data point and all the labeled data points
- (a) i, ii, iii, iv
 - (b) iv, iii, ii, i
 - (c) iv, i, iii, ii
 - (d) ii, iii, i, iv

Question 36. Consider a 3×3 matrix A with rank 1 where $Ax = b$ for some b has three solutions $x = \begin{bmatrix} 1 \\ p \\ 3 \end{bmatrix}$, $\begin{bmatrix} 2 \\ q \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 5 \\ r \\ 1 \end{bmatrix}$. Which among the following vectors belong to the null space of A where $p = 2, q = -3$ and $r = -8$?

(a) $\begin{bmatrix} -12 \\ 40 \\ 8 \end{bmatrix}$

(b) $\begin{bmatrix} -3 \\ 5 \\ 53 \end{bmatrix}$

(c) Data insufficient

(d) $\begin{bmatrix} -15 \\ 17 \\ 45 \end{bmatrix}$

Question 37. Find the F -statistic for the following sample sets generated from the distributions with population variances $\sigma_1^2 = 3$ and $\sigma_2^2 = 3$ respectively.

Sample data 1:

25	25	18	16	26	26
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Sample data 2:

20	23	22	21	18	17
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Question 38. How many numbers between 10 and 10,000 can be formed by using the digits 1, 2, 3, 4, and 5 if no digit is repeated in any number?

Question 39. Two numbers are selected independently at random in the interval $[0, 1]$. You are told that the smaller one is less than $\frac{1}{3}$. The probability that the larger one is greater than $\frac{3}{4}$ is _____.

Question 40. Which of the following measures are robust with respect to outliers?

- (a) Mean
- (b) Standard deviation
- (c) Median
- (d) Variance